Chapter 16
Options on Indices, Currencies, and Futures

Chapter 15 derived the Black–Scholes–Merton formula for options on a non-dividend-paying stock, a small corner of the option market by notional. The largest option markets in the world are written on interest rates, currencies, and futures contracts, and almost none of them involve a share that pays nothing. Currency options are also the first option market in this book that trades mainly over the counter, dealer to client, rather than on an exchange; Section 10.10 set out what that changes.

The extension needed is smaller than it looks. Section 15.9 already stated it: replacing S_0 with $S_0 e^{-qT}$ covers every case, with q interpreted differently in each. This chapter works through what q means in each market, what conventions surround the resulting prices, and one genuinely new structure, the quanto, where an option's payoff is fixed in a currency other than the one its underlying trades in.

It also introduces Black's model, the form the Black–Scholes–Merton formula takes when written on a forward or futures price. That form, not the original, is the market standard almost everywhere outside single-stock equity options, and Chapter 29's caps and swaptions are written in it or in its normal-model variant (Section 29.8).

Section 16.1 states the one substitution. Sections 16.2 to 16.4 apply it to index options and to currency options, including how the FX market actually quotes. Sections 16.5 and 16.6 are Black's model, on futures and beyond them, and Section 16.7 is the quanto. Sections 16.8 and 16.9 close with the two practical questions: where the forward comes from, and which convention each market uses.

16.1 ONE SUBSTITUTION, THREE MARKETS

One observation unifies the chapter: a *yield* on the underlying and a *dividend* on a stock enter the pricing problem identically. Both mean that holding the asset produces income the option holder does not receive, so the asset behaves as though it were worth $S_0 e^{-qT}$ today.

With that substitution, equations (15.2)–(15.4) become

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2) \tag{16.1}$$

$$p = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1) \tag{16.2}$$

$$d_1 = \frac{\ln(S_0/K) + \left(r - q + \sigma^2/2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

Table 16.1 gives the interpretation of q in each market. Nothing else changes. The last row gets a section of its own: Section 16.5 shows that setting $q = r$ gives Black's model.

Table 16.1: What q means in equation (16.1).

Underlying	q	Reason
Non-dividend stock	0	Holding it produces no income
Dividend-paying stock or index	Dividend yield	Holder receives dividends; option holder does not
Currency	Foreign rate r_f	Foreign currency earns interest abroad
Futures contract	r	Costs nothing to hold, so it grows at zero under Q

16.2 OPTIONS ON STOCK INDICES

For an index, q is the dividend yield of the constituent stocks over the option's life: an average, weighted by index weight, and known reasonably well in advance because dividend policies change slowly. Section 15.9 priced a straightforward index call. The more instructive application is the one Section 12.10 introduced without a model: using index puts to insure a portfolio.

Example 16.1 Insuring a portfolio, and why the strike is not obvious

A fund holds \$30 million tracking an index at 1,500, with a dividend yield of 2%. The manager wants a floor at \$28.5 million (a 5% loss) over six months, with $r = 4\%$ and $\sigma = 18\%$.

If the portfolio mirrors the index, the natural instrument is a put struck 5% below the current level, at 1,425. The contract multiplier is \$100, so

$$\text{contracts} = \frac{30{,}000{,}000}{1{,}500 \times 100} = 200$$

and from (16.2) with $K = 1{,}425$, the put is worth 37.27 index points, costing

$$200 \times 100 \times 37.27 = \$745{,}400$$

or 2.48% of the portfolio for six months.

A floor at 1,425 does not produce a floor at \$28.5 million. The portfolio also collects dividends over the six months (about 1% at a 2% annual yield), so at index 1,425 the fund holds roughly $30\text{m} \times 0.95 + 0.30\text{m} = \$28.8$ million before the put pays anything, and about \$28.05 million net of the premium, below the floor that was wanted. Separately, the put's payoff arrives at expiration while the insurance was wanted throughout.

A manager who wants a genuine floor on *terminal wealth* must solve for the strike that delivers it, accounting for dividends received and the premium paid. If the portfolio is not an index tracker its beta enters as well; the next paragraph works that out. The mechanical calculation is easy and the specification of what is being insured is not, which is where most portfolio-insurance mandates go wrong.

When the portfolio is not the index. A portfolio with beta β against the index does not fall 5% when the index does, so the strike that protects it has to be found through the beta. Write V for the portfolio's value (Section 3.6's V_A), and keep S for the index the put is written on. The capital asset pricing model (CAPM) relates the two total returns over the option's life,

$$\frac{V_T}{V_0} + q_V T - e^{rT} = \beta \left(\frac{S_T}{S_0} + qT - e^{rT} \right)$$

and solving for the index level gives

$$\frac{S_T}{S_0} = \frac{1}{\beta} \left(\frac{V_T}{V_0} + q_V T \right) - qT + \left(1 - \frac{1}{\beta} \right) e^{rT} \tag{16.3}$$

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where q_V and q are the dividend yields of the portfolio and the index, and S_T, the index level at which the portfolio reaches its floor, is the put strike. Take the fund of Example 16.1 with $\beta = 1.5$, dividend yields of 2% on both portfolio and index, and $r = 4\%$. A \$28.5 million floor on terminal wealth, dividends included but before the premium, is $V_T/V_0 + q_V T = 0.95$, so

$$\frac{S_T}{S_0} = \frac{0.95}{1.5} - 0.01 + \left(1 - \frac{1}{1.5}\right) e^{0.02} = 0.9634$$

and the puts are struck at $0.9634 \times 1{,}500 = 1{,}445$, not 1,425: a portfolio with beta 1.5 has lost 5%, dividends included, by the time the index level is down only 3.7%. The number of contracts scales with beta as well, $1.5 \times 200 = 300$, exactly as Section 3.6's hedge ratio did. At $\beta = 1$ equation (16.3) reduces to $S_T/S_0 = V_T/V_0 + (q_V - q)T$, the dividend point made in the example.

Example 16.1 bought a put, but through the mid-1980s a large body of institutional money instead manufactured one, a technique marketed as **portfolio insurance**: selling index futures as the market fell, in the proportion a put's delta prescribes (Section 18.11). Section 40.2 tells how that played out on October 19, 1987. The equity skew dates from that day, and Chapter 19's case study "The day the skew appeared" asks why.

Who trades index options

Index options are the largest equity option market by notional, and the participants divide cleanly.

Protection buyers are the users of Example 16.1: pension funds, insurers, and asset managers buying puts or put spreads against a portfolio. Section 12.10 established the arithmetic that governs them: continuous protection struck 5% out of the money costs roughly 5% a year, so almost everyone buys a spread or a collar instead.

Overwriters systematically sell calls against index holdings. Section 12.12's case study on derivative-income funds described the largest such program, and Example 10.7 gave the shape: an uncertain equity return converted into a capped one.

Volatility traders take positions in index volatility itself, delta-hedged as Section 12.4 described. They are the natural counterparty to both groups above, and Part IV treats them.

Dispersion traders exploit the relationship between index volatility and the volatilities of its constituents. An index is less volatile than the average of its members because they are imperfectly correlated (Section 14.11's two-asset volatility in miniature), so selling index volatility and buying single-stock volatility is a position in correlation. Chapter 22 develops it.

The structure of that list explains a persistent feature of the market. Protection buyers and overwriters both trade one way, buying puts and selling calls respectively, and dealers absorb the imbalance. The equity skew is partly the price of that imbalance, which is a market-structure explanation to set alongside the crash-risk explanation that dates the skew to 1987. Chapter 19 weighs them.

16.3 CURRENCY OPTIONS

A foreign currency is an asset that pays interest at the foreign rate, so $q = r_f$. Substituting into (16.1) gives what is often called the Garman–Kohlhagen formula, though it is the same equation:

$$c = S_0 e^{-r_f T} N(d_1) - K e^{-rT} N(d_2) \tag{16.4}$$

with S_0 the spot exchange rate expressed as units of domestic currency per unit of foreign.

That last phrase carries more difficulty than the formula. Every quantity in an FX option must be labeled with which currency it is measured in, and errors of this kind are a common source of loss in currency options among people who understand the mathematics perfectly.

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Example 16.2 A euro call

A US firm expects to pay €1 million in six months and wants protection against a rising euro. Spot is 1.0850 dollars per euro, US rates are 4%, euro rates 2.5%, and volatility is 8%. Value a call on the euro struck at 1.1000.

Here the domestic currency is the dollar, so $r = 4\%$, $q = r_f = 2.5\%$, $S_0 = 1.0850$, $K = 1.1000$.

$$d_1 = \frac{\ln(1.0850/1.1000) + \left(0.04 - 0.025 + 0.08^2/2\right) \times 0.5}{0.08\sqrt{0.5}} = -0.0819, \qquad d_2 = -0.1384$$

$$c = 1.0850e^{-0.0125} \times 0.467383 - 1.1000e^{-0.02} \times 0.444954 = 0.50081 - 0.47976 = 0.02105$$

On a €1 million notional the premium is $21,053.

Check the forward, since it is the reference every FX desk uses:

$$F = 1.0850\, e^{(0.04-0.025)\times 0.5} = 1.09317$$

The strike of 1.1000 is above the forward, so the call is out of the money *forward*, and indeed $d_2 < 0$, a risk-neutral exercise probability of 44.5%.

The premium is in dollars per euro of notional; Table 16.2 sets out the three other ways the same trade is quoted.

Who buys currency options, and what they buy

Currency options are bought mostly by people with a currency exposure they did not choose, and the structures reflect that.

Importers and exporters face the exposure of Example 16.2: a known foreign-currency payment or receipt at a known date. A plain option caps the cost, but the premium is a visible expense that must be defended, and Section 12.10 explained why that matters more than it should. The usual response is a **range forward**, also called a zero-cost collar or cylinder, which Example 16.3 constructs.

Example 16.3 A zero-cost range forward

The importer of Example 16.2 owes €1 million in six months and does not want to pay a premium. They buy a euro call struck at 1.1300 and sell a euro put struck so the premiums offset.

The call at 1.1300 is worth 0.01063 dollars per euro. Solving for the put strike with the same value gives $K_p = 1.0587$, and the structure costs nothing at inception.

The result is a band:

Euro at expiration	What happens	Cost of €1m
Below 1.0587	Put exercised against them	$1,058,748
1.0587 to 1.1300	Both expire; buy at spot	$1,058,748–$1,130,000
Above 1.1300	They exercise the call	$1,130,000

Compare the outright forward, which locks in $1,093,168 with certainty. The collar gives up the chance of paying less than $1,058,748 in exchange for capping the cost at $1,130,000, and costs nothing today.

The trade-off is the one Section 12.10 identified for equity index collars, with one difference. In equities the skew forces an asymmetric band: the put bought is further from spot than the call sold. Here, at a flat 8% volatility, the band is nearly symmetric around the forward (3.1% below it, 3.4% above); with a smile like Figure 16.1's, where euro puts carry the higher volatility, the put the importer sells is worth more and can be struck further out. The shape of the collar a client can be offered is a direct reading of the shape of the smile, which is why Section 16.4's quoting conventions matter commercially and not just descriptively.

Cross-border investors hedge the currency exposure that comes with a foreign asset.

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Options are used rather than forwards when the size of the foreign holding is uncertain: an equity portfolio's value moves, so the required hedge moves with it, and an option's payoff scales in a way a fixed forward does not.

Carry traders buy protection on the funding currency, the low-yield currency borrowed to finance the position. A position long a high-yield currency against a low-yield one earns the differential and loses if the funding currency rallies, which tends to happen abruptly, when many such positions are unwound at once. An out-of-the-money option on the funding currency converts an open-ended tail into a premium, and the demand for it is visible in the risk reversal.

Structured product issuers embed currency options in notes sold to retail investors. The simplest is the dual-currency deposit, in which the investor sells a currency option in exchange for an enhanced deposit rate. Section 12.12's test applies: work out what option the buyer has written, and ask whether they would have written it deliberately.

Macro funds trade currency volatility and the risk reversal outright, as views on an exchange rate rather than hedges of it.

16.4 HOW THE FX MARKET ACTUALLY QUOTES

Currency options are quoted differently from every other option market, and the conventions are worth knowing because Chapter 19 uses them and they are more sensible than they look.

Which currency is the premium in? Table 16.2 sets out the four correct ways to quote the premium of Example 16.2.

Table 16.2: One premium, four quotations. €1m call struck at 1.1000; spot 1.0850.

Convention	Computed as	Value
Domestic per unit foreign	the formula's direct output	0.02105
% of foreign notional	premium / S_0	1.940%
% of domestic notional	premium / K	1.914%
Domestic cash	premium × foreign notional	\$21,053

The first and second differ by whether the premium is paid in dollars or euros, the second and third by which leg's notional is the base, and the fourth is the first multiplied by the notional. Every one is correct and they are not equal, which is why FX confirmations state the premium currency, amount, and settlement date rather than a percentage.

A related trap concerns delta. A euro call is also a dollar put, so a position has two deltas depending on which currency is treated as the asset, and the two are related through the **premium-adjusted delta**: the ordinary delta less the premium expressed in units of the foreign currency, $\Delta - c/S_0$, which is the delta seen by a holder who pays the premium in that currency. Desks additionally distinguish *spot* delta from *forward* delta, and whether the premium has been deducted. The number of ways to be right is large, and the number of ways to be consistent with a counterparty is one.

Volatility, not price. As Section 10.6 described generally, FX options trade on implied volatility. But the FX market goes further: it identifies options by *delta* rather than by strike. A "25-delta put" is the put whose delta is -0.25, whatever strike that turns out to be today.

The reason is that a strike ages badly. What matters for an option's price is how far its strike sits from the forward in standard deviations,

$$\frac{\ln(K/F_0)}{\sigma\sqrt{T}}$$

and for a fixed strike that distance changes every day: the forward moves with spot, and even if it

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did not, the shrinking $\sigma\sqrt{T}$ pushes the same strike further from the money in standard-deviation terms as expiration approaches. A strike set at today's forward is at the money *forward* today (the sense Example 16.2 used, which coincides with the spot sense only when $r = r_f$) and is unlikely to be at the money in either sense next month. A 25-delta option is defined by that distance itself, so it is always the same distance out of the money. Delta is a moneyness measure that is automatically rescaled by volatility and time, and quoting in it keeps a surface comparable across days. Figure 19.4 draws exactly this, marking one strike on the smile before and after a move in spot.

Three quotes per maturity. Rather than a volatility for each strike, the FX market quotes three numbers at each maturity, the at-the-money (ATM) volatility, the risk reversal (RR), and the butterfly (BF), from which the whole smile is reconstructed, as Example 16.4 does:

$$\text{ATM} = \text{volatility of the delta-neutral straddle}$$

$$\text{RR}_{25} = \sigma_{25\text{-delta call}} - \sigma_{25\text{-delta put}} \tag{16.5}$$

$$\text{BF}_{25} = \frac{1}{2}\left(\sigma_{25\text{-delta call}} + \sigma_{25\text{-delta put}}\right) - \text{ATM} \tag{16.6}$$

The delta-neutral straddle is struck close to the forward but not at it; long-dated and some emerging-market pairs use the at-the-money-forward straddle instead. Strictly, the quoted butterfly fixes the price of a strangle valued at the single volatility ATM + BF, and (16.5) and (16.6) are the first-order reading desks use for a quick reconstruction.

The risk reversal quote measures the smile's *skew* (whether calls or puts are more expensive in volatility terms), and Section 12.6 showed it is the instrument that trades exactly that. The butterfly quote measures the smile's *curvature*, how much the wings exceed the middle, and the instrument that isolates it is a strangle bought against a straddle in vega-neutral proportions. This is not a coincidence: the market quotes what it trades, and it trades level, skew, and curvature because those are the three things a smile can do.

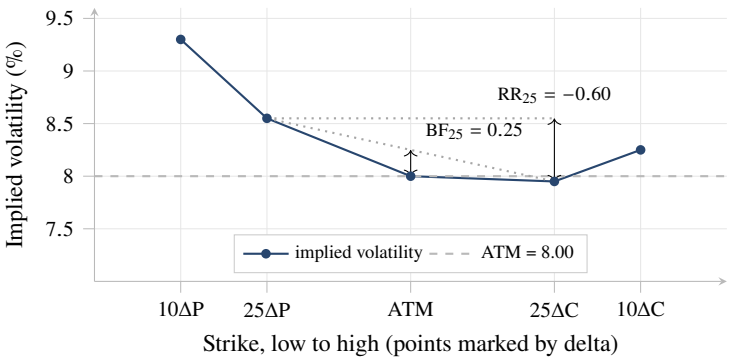


Figure 16.1: Smile built from ATM 8.00, $\text{RR}_{25} = -0.60$, $\text{BF}_{25} = 0.25$; the 10-delta points are extrapolated. Dotted: the 25-delta put level, and the chord joining the two 25-delta points. Illustrative.

Example 16.4 Reconstructing a smile from three numbers

A dealer quotes six-month EUR/USD as ATM 8.00, 25-delta risk reversal −0.60, 25-delta butterfly 0.25.

Solving (16.5) and (16.6) for the two wing volatilities:

$$\sigma_{25\text{C}} = \text{ATM} + \text{BF} + \frac{1}{2}\text{RR} = 8.00 + 0.25 - 0.30 = 7.95\%$$

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$$\sigma_{25P} = \text{ATM} + \text{BF} - \frac{1}{2}\text{RR} = 8.00 + 0.25 + 0.30 = 8.55\%$$

The three quotes fix three points on the smile, the at-the-money and the two 25-delta points of Figure 16.1; the figure's 10-delta points are extrapolated. Adding 10-delta quotes fixes five points, which is what most desks use.

Read the risk reversal. It is negative, meaning euro puts are more expensive than euro calls in volatility terms: the market is paying more for protection against a falling euro. In equity markets the skew has this sign essentially always (Section 15.11); in FX it changes sign as sentiment moves, which makes it a genuinely informative price rather than a structural constant.

16.5 OPTIONS ON FUTURES AND BLACK'S MODEL

The last row of Table 16.1 produces the most widely used version of the formula. A futures contract costs nothing to enter, so in the risk-neutral world its price has zero expected growth; Section 13.10 derived this from the tree, and Section 14.7 from Itô's lemma. Setting $q = r$ in (16.1) makes the $S_0 e^{-qT}$ term equal $F_0 e^{-rT}$, and the formula collapses to

$$c = e^{-rT}\,[F_0 N(d_1) - K N(d_2)] \tag{16.7}$$

$$p = e^{-rT}\,[K N(-d_2) - F_0 N(-d_1)] \tag{16.8}$$

$$d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

This is **Black's model**, and Example 16.5 applies it. Three features make it the market's preferred form.

The interest rate appears only as a discount factor. It does not enter d_1 or d_2 at all. Everything about the underlying is contained in F_0, which is directly observable, so the model requires no view on carry, dividends, convenience yields, or foreign rates, all of which are already embedded in the futures price.

The forward is the natural reference. Setting $K = F_0$ makes $c = p$, which Section 11.4 derived from parity alone. Black's model reproduces it immediately, and this is why the at-the-money-forward straddle is the usual volatility reference point outside FX (Section 16.4).

It generalizes. Anything with an observable forward price can be priced this way, whether or not a futures contract exists; Section 16.6 shows how far that reaches.

Example 16.5 An option on a futures contract

A futures price is 100, with $r = 4\%$, $\sigma = 25\%$, and six months to expiration. Value calls and puts struck at 100.

$$d_1 = \frac{\ln(100/100) + 0.25^2 \times 0.5/2}{0.25\sqrt{0.5}} = 0.0884, \qquad d_2 = -0.0884$$

$$c = e^{-0.02}\,[100 \times 0.535216 - 100 \times 0.464784] = 6.904$$

and the put comes to the same 6.904, as it must at $K = F_0$.

As a consistency check, note that for a non-dividend stock $F_0 = S_0 e^{rT}$, so Black's model on the futures price and the Black–Scholes–Merton formula on the spot must agree. Taking Example 15.1's option ($S_0 = 58$, $K = 55$, $r = 4\%$, $\sigma = 24\%$, $T = 0.5$) gives $F_0 = 58e^{0.02} = 59.1717$, and (16.7) returns \$6.1731, identical to four decimals. The two are the same formula written in different variables.

How exercise settles. Exercise does not deliver the commodity, and it does not happen at an intraday price. At CME exercise notices are processed after the close: the holder of a call is given a long futures position opened at the strike, and that position is marked to the exercise

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day's settlement price like any other. The first mark-to-market therefore pays the holder that settlement price less the strike, and leaves a long futures position at the settlement price. The writer, on assignment, pays the cash and takes the short position.

Together the cash and the position are a futures contract bought at K: the cash is the settlement price less K, and the position, worth nothing once marked, earns only the moves after the close, so it is a new futures position rather than part of the payoff. The book therefore takes the payoff as $\max(F - K, 0)$ with F the settlement price on the day of exercise. The price at which the holder decided to exercise does not enter: the move from it to the settlement is added to a call's payoff and subtracted from a put's, which is one reason many contracts now settle straight to cash. Example 16.6 works through a call and a put.

Example 16.6 Exercising a futures call, and a futures put

A trader holds one call on the December crude oil futures contract struck at \$70.00, on 1,000 barrels. During the day, with the futures at \$73.40, the call's intrinsic value is

$$1{,}000 \times (73.40 - 70.00) = \$3{,}400,$$

and the trader submits an exercise notice. The contract settles that evening at \$73.85, and the notice is processed after the close: a long futures position opened at \$70.00 and marked to \$73.85. The move after the decision adds

$$1{,}000 \times (73.85 - 73.40) = \$450$$

for a total of \$3,850, which is $1{,}000 \times (73.85 - 70.00)$: the exercise day's settlement price less the strike, exactly as (16.7) assumes.

The put is the mirror image. A put on the same contract struck at \$76.00, exercised at the same moment, is worth

$$1{,}000 \times (76.00 - 73.40) = \$2{,}600$$

at \$73.40, but it opens a *short* futures position at \$76.00, marked to \$73.85. The same \$450 the call gained is lost, leaving \$2,150, which is $1{,}000 \times (76.00 - 73.85)$. The move from the intraday price to the settlement price helps one and hurts the other by the same amount, and both holders end with the intrinsic value at the settlement price.

Keeping the futures position leaves the cash and an outright position at \$73.85, long for the call holder and short for the put holder. That is a bet on crude, not a larger payoff, and it is the choice an assigned writer faces in reverse.

One settlement feature, not universal, does change the value. Where the exchange margins the option itself (**futures-style margining**, used at Eurex and ICE Futures Europe), the premium is not paid up front but posted and marked daily, so there is nothing to discount and Black's model loses its discount factor:

$$c = F_0 N(d_1) - K N(d_2)$$

At the rates of the 1980s the missing discount factor was a large difference; at recent rates it is small but not zero. Exercising early would also release nothing, so an American futures-style option is worth its European value.

Futures options versus spot options. Where both exist, they are not interchangeable, for four reasons, the first of which Section 10.3 gave.

First, exercise delivers a margined position, not an asset to fund. That changes early exercise, so the agreement checked in Example 16.5 holds only for European options. Example 13.8 showed why an American call on a futures contract can be exercised early with no dividend anywhere in the problem. A futures contract costs nothing to hold, which is exactly what $q = r$ encodes, so the premise of Section 11.8's argument, that deferring the strike payment earns

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interest, fails; and exercise releases the intrinsic value as margin that can earn interest. Two practical consequences follow. Listed commodity and rate options with the premium paid up front are usually American, so Black's model gives a lower bound rather than the value, and desks price them on lattices as Section 15.10 described. And a writer of American futures options should expect assignment whenever the option is deep in the money, not merely near expiration.

Second, where the futures contract expires with the option, a European futures option and a European spot option are worth the same. American ones are not: when the futures price stands above spot (contango), the American call on the futures is worth more than the American call on the spot and the American put less, and in backwardation the ranking reverses.

Third, the underlying futures contract usually expires shortly after the option, so the option's underlying is a specific contract month rather than the spot. On a commodity with a steep curve, that difference is material.

Fourth, the futures price already contains the carry. Where the carry is uncertain (an unknown convenience yield, an unpredictable dividend), pricing off the futures removes an estimation problem that the spot-based formula would have to solve.

Who uses them. Section 10.3 named the hedgers; a refiner buying crude calls or a grower buying corn puts gets a cap or a floor where a futures hedge would be a lock. Asset managers and mortgage servicers use options on Treasury and rate futures to hedge duration and the convexity of Chapter 26's prepayment risk. By notional, the largest listed rate options are those on SOFR futures, which absorbed the Eurodollar options in 2023; since the futures price is 100 less the rate (Section 6.1), a call on the price is a put on the rate, which is why a SOFR call hedges a fall in rates. Exchanges also list weekly and daily expirations on equity index, Treasury, crude oil and gold futures, the futures-market counterpart of Chapter 23's short-dated options. Commodity trading advisors and volatility funds trade them for the volatility itself, and they are the natural counterparty to the hedgers.

Where commodity and energy options live

Black's model is not a special case in commodities—it is the whole market. Almost every listed option on oil, gas, metals, or grain is written on a *futures contract* rather than on the physical, and there are good reasons.

The spot price may not exist in a usable form. Section 2.1's contract specification showed that a commodity's price is a location, a grade, and a delivery window, not a single number, and Section 36.9 develops the point. A futures contract fixes all three, so it is the only unambiguous price to write an option on.

Delivery is avoided. Exercising a futures option produces a futures position, which can be closed. Exercising a physical option produces an obligation to take delivery of crude oil, which most option users are not equipped to do.

Wherever the underlying is awkward to hold, the market writes the option on the forward and prices it with Black's model; Section 16.6 carries this to interest rates.

16.6 BLACK'S MODEL BEYOND FUTURES

Nothing in (16.7) requires a futures contract. All it needs is a forward price that is observable and that has zero expected growth under the measure used for discounting. That condition holds far beyond the futures market, and the world's largest option market runs on it.

A caplet is a call on a forward interest rate: Black's model applies with F_0 the forward rate and K the cap rate, scaled by the accrual fraction and discounted to the payment date at the end of the period. A swaption is an option on a forward swap rate, and the same form applies with the discount factor replaced by the *annuity*, the present value of one unit paid on each

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fixed-leg date (Chapter 7's PV01 leg). Chapter 29 derives both formulas and the normal-model form in which the market now mostly quotes them (Section 29.8), and covers who trades them: borrowers capping floating funding, issuers of callable debt, and pension funds hedging a fall in rates.

The condition that must be checked. Black's model requires the forward to have zero drift under the discounting measure, and this is where the apparatus can be misapplied. A forward rate is a martingale under the measure associated with *its own* payment date. A forward swap rate is a martingale under the measure associated with *the annuity*. Use one where the other belongs (pay a rate at the wrong date, observe it in the wrong currency, or average it over a period) and the forward acquires a drift.

That drift is exactly the **convexity, timing, and quanto adjustments** of Chapter 30, and Section 16.7 shows the simplest case next. Black's model is not wrong in those situations; it is being fed the wrong forward, and the adjustment corrects the forward rather than the formula.

16.7 QUANTOS AND COMPOSITE OPTIONS

Only one structure in this chapter needs new machinery: an option on a foreign asset whose payoff is settled in the domestic currency. Two versions exist, and confusing them is expensive, as Example 16.8 shows.

A **composite** (or "compo") option pays the foreign-currency payoff converted at the *prevailing* exchange rate at expiration. The holder is exposed to both the asset and the currency, and the option is effectively written on the asset measured in domestic currency. Write V for the asset's foreign-currency price and S for the exchange rate in domestic units per foreign unit, as in Section 16.3. The option's volatility is that of the product VS, which by Section 14.11 is

$$\sigma_{\text{compo}} = \sqrt{\sigma_V^2 + \sigma_S^2 + 2\rho\sigma_V\sigma_S}$$

where σ_V and σ_S are the volatilities of the asset and of the exchange rate, and ρ is their correlation.

A **quanto** pays the foreign-currency payoff converted at a *fixed* rate agreed in advance, typically 1.0, so that a foreign index moving 100 points pays 100 domestic currency units. The holder has no currency exposure at all, which is what makes quantos popular: an investor gets foreign equity exposure without an FX position. CME's dollar-denominated Nikkei 225 futures contract is a listed quanto; over the counter, structured-note issuers are major buyers of quanto calls, and the dealers who sell them carry the correlation risk described below. That convenience has a price, and the price is not obvious.

The quanto adjustment. The issuer of a quanto must hedge by holding the foreign asset, and the size of that foreign holding, measured in domestic currency, changes as the exchange rate moves. If the asset and the currency are correlated, the hedge must be adjusted in a way that is systematically profitable or costly, and the option's price must reflect it.

The result, derived in Chapter 30, is that the foreign asset's forward price must be adjusted:

$$F_0^{\text{quanto}} = V_0\, e^{(r_f - q_f - \rho\,\sigma_V\sigma_S)T} \tag{16.9}$$

where V_0 is the asset's foreign-currency price today, r_f the foreign rate, and q_f its dividend yield. The option is then priced with Black's model (16.7), discounting at the domestic rate.

The correction term $-\rho\sigma_V\sigma_S$ is the **quanto adjustment**. Its sign follows the correlation: when the foreign asset and the foreign currency move together, the adjustment reduces the forward and cheapens a call.

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Example 16.7 Pricing a quanto call

A foreign index stands at 3,000 with a 2% dividend yield. Foreign rates are 1%, domestic rates 4%. Index volatility is 20%, exchange-rate volatility 10%, and their correlation is +0.3. Value a one-year quanto call struck at 3,100, paying one domestic unit per index point.

From (16.9),

$$F_0^{\text{quanto}} = 3000\, e^{(0.01-0.02-0.3\times0.20\times0.10)\times1} = 3000\, e^{-0.016} = 2{,}952.38$$

Without the correction the forward would be $3000e^{-0.01} = 2{,}970.15$, so the quanto adjustment lowers the forward by 17.77 points, or 0.60%.

Applying Black's model with $\sigma = 20\%$ and discounting at the domestic 4%:

$$d_1 = \frac{\ln(2{,}952.38/3{,}100) + 0.20^2 \times 1/2}{0.20\sqrt{1}} = -0.1439, \qquad d_2 = -0.3439, \qquad c = 167.52$$

against 175.18 if the adjustment were omitted: an error of 4.6% in the option's value from a 0.6% error in the forward. Options magnify errors in the forward, so the adjustment cannot be waved away as second order.

Two practical points follow. The price depends on ρ, a correlation that is not directly traded and must be estimated, so quantos carry a risk that vanilla options do not. That risk is why quanto books carry **correlation reserves**: part of the book's model value held back against the chance that the correlation has been misestimated. And the sign matters: had the correlation been −0.3, the adjustment would have *raised* the forward and the call.

Example 16.8 The same view, two structures

Take the market of Example 16.7 and price the composite alongside the quanto. Assume the exchange rate starts at 1.00 domestic unit per foreign unit, so both structures start from an index level of 3,000 in domestic terms, and both are struck at 3,100 for one year.

Composite. The underlying is the index measured in domestic currency, whose volatility is

$$\sigma_{\text{compo}} = \sqrt{0.20^2 + 0.10^2 + 2\times0.3\times0.20\times0.10} = 24.90\%$$

Its forward is $3000e^{(r-q_f)T} = 3000e^{0.02} = 3{,}060.60$, and Black's model gives

$$c_{\text{compo}} = 274.69$$

Quanto. From Example 16.7, $c_{\text{quanto}} = 167.52$.

The composite costs 64% more, for two reasons: it carries a higher volatility (24.9% against 20%), and its forward is higher, 3,060.60 against 2,952.38. Most of the gap between the forwards is the foreign currency's own forward premium, since the domestic rate exceeds the foreign rate by three percentage points; the remaining 0.6% is the quanto adjustment the composite escapes.

The payoffs explain the price gap better than the formulas do:

Index	FX rate	Quanto pays	Composite pays
3,400	0.90	300	0
3,400	1.00	300	300
3,400	1.10	300	640
3,100	1.10	0	310
2,800	0.90–1.10	0	0

The quanto pays on the index alone. The composite pays on the index *and* the currency, so it can pay when the index ends at the strike and the currency has risen—and pay nothing when the index has risen and the currency has fallen.

These are different views, not different prices for the same view. An investor who wants foreign equity exposure and no currency exposure buys the quanto and pays for the correlation adjustment. An investor who is bullish on both buys the composite and pays for the extra volatility. Selling one

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to a client who wanted the other is an unsuitable sale that the payoff table makes obvious and the price comparison does not.

16.8 GETTING THE FORWARD RIGHT

Every formula in this chapter takes a forward as its real input. Black's model says so explicitly; equation (16.1) says so in disguise, since $S_0 e^{-qT}$ is the discounted forward $F_0 e^{-rT}$. The practical question in each market is not which formula to use but where the forward comes from. Table 16.3 gives the answer, and one rule governs all of it.

Table 16.3: Where F_0 comes from in each market.

Market	Computed as	But in practice
Equity index	$S_0 e^{(r-q)T}$	Read from index futures
Single stock	$S_0 e^{rT}$ − FV(divs)	Implied from the option chain
Currency	$S_0 e^{(r-r_f)T}$	Read from the FX forward market
Commodity	Not computable	The futures price
Interest rate	From the curve	From the curve (Ch. 4)

The rule is to use the traded forward wherever one exists, never the computed one. Earlier chapters gave the reasons, and they are not small.

For currencies, Section 5.7 showed that covered interest parity has been persistently violated since 2008, by tens of basis points in the deepest markets in existence. A forward computed from $S_0 e^{(r-r_f)T}$ will therefore differ from the forward one can actually trade at, and the difference is the cross-currency basis of Chapter 7. Pricing an FX option off the computed forward embeds an error that counterparties can trade against. For commodities the forward is not computable even in principle (Section 5.8); the futures price is the primary datum. For single stocks, Section 11.6 showed the implied forward can be read directly from the option chain; Example 16.9 does the same for an index and shows what that forward contains.

Example 16.9 Extracting the dividend yield from index options

An index stands at 1,500. Six-month options struck at 1,500 trade at 68.50 (call) and 61.20 (put), with $r = 4\%$. What dividend yield is the market assuming?

Put–call parity with a yield, from Section 11.4, requires

$$c - p = S_0 e^{-qT} - K e^{-rT}$$

Substituting,

$$7.30 = 1500\, e^{-0.5q} - 1500\, e^{-0.02} = 1500\, e^{-0.5q} - 1470.30$$

so $1500 e^{-0.5q} = 1477.60$, giving

$$e^{-0.5q} = 0.985067 \quad \Longrightarrow \quad q = 3.01\%$$

Three points follow, and the third is the one that matters. The calculation needs no volatility and no model: it is parity, which Chapter 11 established without either. It gives the market's dividend expectation, which for an index is often better than a published forecast, because index dividends are actively traded.

And the number is not purely a dividend yield. It is whatever makes the forward come out right, which bundles expected dividends with the cost of financing or borrowing the index basket. What has been measured is the implied forward, expressed as a yield: the same warning Section 11.6 gave for single stocks. A desk that treats 3.01% as a dividend forecast and separately adds a financing spread has counted the same thing twice.

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16.9 WHICH CONVENTION FOR WHICH MARKET

Table 16.4 collects what a reader needs to move between markets without importing the wrong assumption.

Table 16.4: Pricing form and quoting convention by market.

Market	Form used	Quoted in	Moneyness by
Single-stock equity	Black–Scholes–Merton, discrete dividends	Price	Strike
Equity index	Black–Scholes–Merton with q	Price or volatility	Strike
Currency	Black–Scholes–Merton with r_f	Volatility	Delta
Commodity, energy	Black's model on futures	Volatility	Strike or delta
Interest rate (caps, swaptions)	Black or normal model on forward rates	Volatility	Strike

Two habits follow from Table 16.4.

Ask what the forward is before asking what the option is worth. In every row, the model takes a forward as its input, and errors in the forward are magnified in the option; Example 16.7 put the magnification at roughly eight to one. Most disputes about an option's value turn out to be disputes about the forward.

Ask which currency each number is in. Table 16.2 showed one premium admitting four correct expressions. The same discipline applies to notionals, strikes, and deltas in any cross-currency structure.

Why volatilities differ so much between markets

A reader moving between the markets of Table 16.4 meets volatility numbers that differ by an order of magnitude, and the differences are structural rather than accidental.

Table 16.5: Indicative implied volatility ranges.

Market	Typical range
Major currency pairs	5–12%
Emerging-market currencies	8–25%
Equity indices	12–25% calm, 30–80% in crisis
Single stocks	20–60%
Commodities	20–60%, energy higher

The most striking entry in Table 16.5 is the first, and Section 14.11 already explains it. An exchange rate is a *ratio* of two currencies, and by equation (14.19) the volatility of a ratio is

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$$

Two large economies share most of their shocks (global growth, commodity prices, risk appetite), so ρ is high and the common component largely cancels. What remains in an exchange rate is the difference between two economies, not the level of either, and differences are smaller than levels.

Equities sit at the other end for the opposite reason. A share is a residual claim on a leveraged balance sheet, so the volatility of the assets is amplified by the leverage before it reaches the equity. Single stocks exceed indices because diversification removes the idiosyncratic part but not the leverage.

The practical use of Table 16.5 is as a sanity check. A quoted volatility outside its market's range is far more often a units or convention error than a discovery, and Section 14.6's rule of

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thumb (daily move $\approx \sigma/16$) converts any of these into something checkable against a screen in seconds.

SUMMARY

Replacing S_0 with $S_0 e^{-qT}$ covers indices, currencies, and futures, with q the dividend yield, the foreign rate, and the domestic rate. Nothing else in the formula changes.

Portfolio insurance is mechanically simple and conceptually not: a floor on the index level is not a floor on terminal wealth, because dividends are received, premium is paid, and a portfolio with beta other than one moves further than the index.

In currencies the difficulty is not the formula but the labeling: a premium can be correctly quoted four ways, and every quantity must carry its currency. The market quotes volatility rather than price and identifies options by delta rather than strike, because a strike ages and a delta does not. Each maturity is three numbers (ATM, 25-delta risk reversal, 25-delta butterfly) measuring the smile's level, skew, and curvature, which are what the corresponding instruments trade. Unlike equity skew, the FX risk reversal changes sign.

Setting $q = r$ gives Black's model, $c = e^{-rT}[F_0 N(d_1) - K N(d_2)]$: the rate is only a discount factor, and everything about the underlying sits in the observable forward. At $K = F_0$ the call and put are equal. It prices anything with an observable forward, caplets and swaptions included, provided that forward has no drift under the discounting measure; where it does, Chapter 30's adjustments correct the forward, not the formula. Every formula here takes a forward as its real input, and the traded forward is to be used wherever one exists: covered interest parity fails, commodity carry is unobservable, and an implied dividend yield bundles financing with dividends.

Exercising a futures option opens a futures position at the strike, marked to the exercise day's settlement price, so the holder collects the intrinsic value at that settlement price and is left holding the futures. Early exercise is possible without dividends, and the underlying is a specific contract month.

A composite option converts the foreign payoff at the prevailing rate and carries volatility $\sqrt{\sigma_V^2 + \sigma_S^2 + 2\rho\sigma_V\sigma_S}$; a quanto converts at a fixed rate and removes the currency exposure, at the cost of a $-\rho\sigma_V\sigma_S$ correction to the foreign forward's drift. Even a small correction to the drift moves the price materially, and the dependence on an untraded correlation is why quanto books carry correlation reserves.

FURTHER READING

Merton (1973) gives the continuous-yield extension (16.1), Garman and Kohlhagen (1983) is the standard citation for the currency-option form (16.4), and Black (1976) introduces the futures form. On FX quoting conventions (premium currency, spot or forward delta, premium adjustment, and which straddle is at the money), Reiswich and Wystup (2010) set them out in one paper, and Clark's *Foreign Exchange Option Pricing* (2011) works them through as a desk uses them. Chapter 30 derives the quanto adjustment.

PRACTICE QUESTIONS

16.1 An index stands at 800 with a dividend yield of 3%. Value a four-month European call struck at 820 with $r = 4\%$ and $\sigma = 22\%$.

16.2 Explain why $q = r$ for a futures contract, and state which chapters established this in discrete and continuous time.

16.3 Spot GBP/USD is 1.2600, US rates are 4%, UK rates 4.5%, and six-month volatility is

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9%. Value a call on sterling struck at 1.2800, and express the premium both in dollars per pound and as a percentage of the sterling notional.

16.4 A dealer quotes three-month USD/JPY as ATM 9.50, $\mathrm{RR}_{25} = +1.20$, $\mathrm{BF}_{25} = 0.30$. Compute the two 25-delta wing volatilities and state which currency the market is paying more to protect against.

16.5 Using Black's model, value a six-month call on a futures price of 60 struck at 58, with $r = 3\%$ and $\sigma = 30\%$.

16.6 Show that at $K = F_0$ Black's model gives $c = p$, and identify the result from Chapter 11 that this reproduces.

16.7 Explain the difference between a quanto and a composite option, and state which one an investor wanting foreign equity exposure without currency risk should buy.

16.8 A foreign index at 2,500 has $q_f = 1.5\%$, $r_f = 2\%$, $\sigma_V = 22\%$, $\sigma_S = 9\%$, and $\rho = -0.25$. Compute the one-year quanto forward and state whether the adjustment raises or lowers it.

16.9 Every formula in this chapter takes a forward as its real input. Using Table 16.3, state the rule that governs how the forward is obtained for an equity index, a currency, and a commodity, and explain why a desk that computes the forward from spot and a carry rate can be wrong even when its rates are right.

16.10 Using Table 16.4, state which convention applies to an equity index option, a currency option, and an option on a futures contract. Explain what goes wrong when a convention is carried across from the market it belongs to, and give one case where two conventions would both be defensible.

FURTHER QUESTIONS

16.11 Example 16.1 notes that a floor on the index level is not a floor on terminal wealth. Set up the equation that determines the strike delivering a genuine \$28.5 million floor, accounting for dividends received and premium paid, and explain why the answer depends on whether the premium is funded from the portfolio or from outside it.

16.12 The FX market quotes by delta rather than strike. Construct the argument for why this makes a volatility surface more stable across days, then identify a circumstance in which it makes the surface *less* stable and state what desks do about it.

16.13 Example 16.7 found that a 0.60% adjustment to the forward changed the option value by 4.6%. Derive the general relationship between a proportional error in the forward and the resulting proportional error in an at-the-money option's value, and explain why the magnification is largest for short-dated options.

16.14 A trader observes that Black's model on a stock index futures price and the Black–Scholes–Merton formula on the index spot give different answers. List the possible causes in order of likelihood, and state what data you would examine to distinguish them.

16.15 Quanto pricing depends on a correlation that is not directly traded. Explain what this implies for how a quanto book should be risk-managed, what a correlation reserve is intended to cover, and why the problem is worse for long-dated quantos than for short-dated ones.

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